Coding Practices Level 3

Hints and Tips

Unit 3

Contents

Hints and Tips Unit 3

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 3: Implementation Of Coding

Section 1 – Understand tools used in coding

Q1. Identify tools used in coding.

Give a short statement for different tools that are used in coding. Try to add a little detail, just one sentence will be sufficient for each tool. You should include at least 3 different tools.

Q2. Select a suitable development environment that supports two or more programming languages. Within your answer provide reasoning for your choice and consider:

- compatibility
- productivity
- extensibility
- performance

The development environment that you choose must be fit for purpose and be applicable for different languages. You must provide adequate reasoning for the selection of the tool chosen and include each of the bullet points within your answer.

Q3. Describe what is meant by source code control. Within your answer include:

- a description of the key common features of most Source Control Management (SCM) solutions
- the impact of not using SCM
- SCM strategy
- why source code control is important within software development

This will be a detailed description to include all of the bullet points within the question. You should show knowledge of the reasons why source code control is important within software development. You should also demonstrate a basic knowledge of at least one SCM strategy and identify this within your answer.

Q4. Select and evaluate a source code control solution. Consider the following in your answer:

- features
- popularity
- hosting
- fit for purpose
- ease of use

As an evaluation is required in this answer, you must consider the pros and cons of the solution that you select. Within your evaluation you should also comment on each of the bullet points within the question.

Section 2 – Understand key concepts of coding

Q1. Explain the following common coding concepts:

- data
- selection
- sequences
- iterations

Each response in the table will need to be sufficiently detailed to fully explain each coding concept. You could give examples of each one to support your answer.

Q2. Explain the following OOD (object-oriented design) principles:

- encapsulation
- abstraction
- inheritance
- polymorphism

Each response in the table will need to be sufficiently detailed to fully explain each principle. You could give examples of each one to support your answer.

Q3. Explain each of the following five OOD (object-oriented design) concepts:

- classes
- interfaces
- objects (including loose and tight coupling)
- attributes (fields)
- properties

Each response in the table will need to be sufficiently detailed to fully explain each concept. You could give examples of each one to support your answer.

Q4. Explain the following for the Methods OOD (object-oriented) programming concept:

- instantiation
- types of constructors
- Get and Sat methods
- behavioural methods

Each response in the table will need to be sufficiently detailed to fully explain each method
You could give examples of each one to support your answer.

Section 3 - Be able to debug and write code

Q1. Identify how to write and debug interpreted code. Consider the following in your answer:

- how the code is written and structured
- the role of an interpreter
- the steps taken when the code is executed and how the code can be debugged

As an identify question this can be a shorter answer with just a short statement for each bullet point listed. Continue to write in paragraph format and don't be tempted to answer with bullet points.

Q2. Identify how to write and debug compiled code. Consider the following in your answer:

- how the code is written and structured
- the role of a compiler
- the steps taken before the code can be executed
- how the code can be debugged
- why there is a necessity of compilation for some languages

This is another identify question and therefore, similar to question 1 a short statement covering each bullet point will be sufficient. As with the guidance for question 1, answer in paragraph format.

Q3. Identify issues that can occur cross-platform

Provide some detail for the issues that you identify in your answer and don't simply list a title of each issue.

Q4. You've been given a task for this question on page 2, session 3 of the course content (see Time to Practice). To complete the task, you will need to create two coding solutions and upload the following files with your assessment:

- .cs and .exe files for the compiled solution
- .py file for the interpreted solution

In the course material you have been provided with an instructional video for how to write interpreted code (Python) and compiled code (C#). Using the examples provided in the video, you will need to carry out the assessment task, which you can find under Time to practice, further down the page.

Once you have completed the task, you will have .exe, .cs and .py files, which you will need to upload with your assessment for your tutor to check.

Q5. Distinguish the differences between the coding solutions from question 4 above Within your answer, demonstrate understanding of the advantages and disadvantages of compilation against interpretation.

Now you have completed the task in question 4, you need to distinguish the differences between the two types of coding solution. It's important that you include some advantages and disadvantages of compilation against interpretation within your answer.